Dr. Carlos Cifuentes

Scientific Data Analysis · Processing & Validation · Python · SQL

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Profile

Physicist with a PhD in Astrophysics and **8+ years of experience in Python-based scientific data processing**. Experienced in developing **workflows for data integration, validation and quality control**, complemented by training in **applied machine learning and predictive modelling**. I also bring nearly two years of industry experience coordinating technical requirements with clients and software teams. I am motivated by applying scientific rigour and analytical problem-solving to projects with meaningful real-world impact. Available in the Heidelberg area.

Technical skills

Coding — **Python** • NumPy • Pandas • SciPy • Astropy.
Basic knowledge of Java; academic use of C/C++, MATLAB and Mathematica.

Machine Learning — **Scikit-learn** • Supervised learning and model evaluation.

Databases — **SQL / PostgreSQL** • Data validation • Data integrity checks.

Versioning — **Git** and GitHub • Collaborative development.

Operating Systems — **Linux / UNIX** • Shell scripting • macOS.

Statistics — Statistical modelling • Time-series analysis • Uncertainty quantification • R.

Documentation — Scientific reporting • Markdown • LaTeX • MS Excel.

Recent training

- IBM DevOps and Software Engineering Professional Certificate (In progress)
- Coursework in Python/Flask, Docker, Kubernetes and OpenShift (Completed)

Relevant experience

2024–2026 **Postdoctoral Researcher** – Scientific Data Analyst
Centro de Astrobiología (CAB), Madrid

- Developed a **Python pipeline** to estimate temperature and metal content of stars from high-resolution spectra ↗. It uses a new approach based on luminosity to reduce degeneracy between parameters in the spectral fit. The **KOBE planet-search project uses these results** to characterise the stars.
- Investigated why some stellar observations lacked a reliable correction for drifts in the instrument. **Recomputed missing corrections** and recovered 57 usable spectra from 16 hours of telescope time for analysis. Contributed to tests that identified night-to-night calibration differences and **informed a revised calibration strategy**.

2018–2023 **Research Scientist (PhD)** – *Astrophysics*
Centro de Astrobiología (CAB), Madrid | Max Planck Institute für Astronomie (MPIA), Heidelberg

- Turned thousands of photometric measurements into **reliable statistical relations** widely used to characterise stars in planet-search investigations ↗.

- Conducted a **statistical analysis of 2,200+ stars** using the large-scale data release from ESA's Gaia space mission, producing **public reference data** for studies of stellar companions ↗.
- Used these data products to train and compare models in a **machine-learning application** that estimates stellar type from brightness measurements ↗.

2014–2016 **Technical Consultant**
Indra Sistemas (Minsait, Smart Energy), Madrid

- Gathered client requirements for energy monitoring and coordinated device specifications with suppliers and Indra's software team, supporting data integration into the **Sofia2 IoT platform**.
- **Analysed energy use and prepared client reports** to guide energy-management decisions.
- **Demonstrated monitoring solutions to clients** and supported their use after deployment, contributing from initial requirements through to day-to-day operation ↗.

Project leadership & communication

- Led **two analytical research projects**, from proposal preparation and resource planning through data acquisition, validation, analysis and delivery.
- Secured **10 competitive observing programmes as Principal Investigator**, coordinating technical requirements, schedules and instrumentation resources.
- Prepared technical reports and scientific documentation for multidisciplinary audiences, delivering **35+ specialist presentations**.
- Contributed to **126 scientific publications**, including **9** first-, second- or third-author papers; **>2800 citations**, **h-index: 30**.

Education & additional experience

- *PhD* in **Astrophysics** (UCM, 2023), Highest Distinction • *MSc* in **Astrophysics** (UCM, 2017) • *MSc* in **Renewable Energies** (UAM, 2014) • *BSc/Licentiate* in **Physics** (USAL, 2013).
- **University Lecturer**: designed and taught full courses in *Differential Geometry* (UNIE, 2024) and *Introduction to Astronomy* (Saint Louis University, 2024), including all teaching materials and student assessments.
- **Research Trainee**: wrote code to estimate stellar luminosities from photometry using numerical integration (UCM, 2017); Made coatings for solar heat collectors that absorbed up to 94% of sunlight, tested how heating affected their performance and durability (CIEMAT, 2014).
- **Languages:** Spanish (native) • English (C1) • German (A1).

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1 Research publications

Note: My personal contribution is indicated by the following tags:

SP – Stellar parameters (spectral synthesis, SED fitting) and multiplicity (CCF analysis).
RV – Radial-velocity time-series analysis and orbital modelling of stellar and planetary systems.
DA – Data acquisition, pipeline development, statistical analysis, and scientific visualisation.
WR – Manuscript writing, editing, and literature review.
Gaia – Articles fundamentally based on *Gaia* astrometric, photometric, or RV data products.

1.1 Selected publications

The following publications highlight first-author work, central contributions to the analysis or results, and particularly relevant papers within my research profile.

[10] **C. Cifuentes**, J. A. Caballero, J. González-Payo, *et al.* *Gaia*
The CARMENES input catalogue of M dwarfs. IX. Multiplicity from close spectroscopic binaries to ultra-wide systems. *A&A*, 693, A228 (2025) 33 citations

[9] **C. Cifuentes**, J. A. Caballero, M. Cortés-Contreras, *et al.* *Gaia*
The CARMENES input catalogue of M dwarfs. V. Luminosities, colours, and spectral energy distributions. *A&A*, 642, A115 (2020) 183 citations

[8] **C. Cifuentes**, J. A. Caballero, S. Agustí *Gaia*
One Is the Loneliest Number: Multiplicity in Cool Dwarfs
RNAAS , 5, 129 (2021) 9 citations

[7] J. Lillo-Box, **C. Cifuentes**, O. Balsalobre-Ruza *et al.* SP WR
CAFE follow-up of TESS hot Jupiter candidates left behind: I. Five newly confirmed planets and a false positive
In press in A&A

[6] A. Schweitzer, V. M. Passeger, **C. Cifuentes** SP DA WR *et al.*
The CARMENES search for exoplanets around M dwarfs. Radii and masses of the target stars. *A&A*, 625, A68 (2019) 201 citations

[5] J. González-Payo, J. A. Caballero, **C. Cifuentes** SP DA WR *Gaia* *et al.*
Characterisation of all multiple systems within 10 pc. *MNRAS*, 549, 1 (2026) New!

[4] H. Martínez-Rodríguez, J. A. Caballero, **C. Cifuentes** SP DA WR *Gaia* *et al.*
Exomoons in the habitable zones of M dwarfs. *ApJ*, 887, 2, 261 (2019) 39 citations

[3] M. Cortés-Contreras, J. A. Caballero, *et al.* incl. **C. Cifuentes** DA WR *Gaia* (6/18)
CARMENES input catalogue of M dwarfs. VIII. Kinematics in the solar neighbourhood. *A&A*, 692, A206 (2024) 21 citations

[2] O. Balsalobre-Ruza, J. Lillo-Box, A. M. Silva, *et al.* incl. **C. Cifuentes** RV DA WR (7/37)
KOBE-1: The first planetary system from the KOBE survey. Two planets likely in the sub-Neptune mass regime around a late K-dwarf. *A&A*, 694, A15 (2025) 7 citations

[1] J. A. Caballero, E. González-Álvarez, M. Brady, *et al.* incl. **C. Cifuentes** RV DA WR (7/67)

A detailed analysis of the Gl 486 planetary system. A&A, 665, A120 (2022) 50 citations

1.2 Publications in coauthorship

[54] A. Quirrenbach, P. Amado, J. A. Caballero *et al.* incl. **C. Cifuentes** SP WR (20/127)
Ten years of exoplanet discoveries and stellar astrophysics with CARMENES
Submitted to A&A

[53] M. Perger, A. Psaridi, S. Dreizler, *et al.* incl. **C. Cifuentes** SP WR (8/25)
The CARMENES search for exoplanets around M dwarfs. Two planets orbiting the M dwarf TOI-2285 that probe the radius valley.
Submitted to A&A

[52] J. González-Payo, **C. Cifuentes**, J. A. Caballero SP DA
The Binding-energy Continuum: From Exoplanets to Ultra-wide Binaries
RNAAS, 10, 6.

[51] R. Varas, Morello, M. Zechmeister, *et al.* incl. **C. Cifuentes** SP WR (8/21)
The CARMENES M-dwarf projected rotational velocity catalogue: a homogeneous analysis accounting for limb darkening
A&A, 709, A244.

[50] T. Poultourtzidis, G. Lacedelli, E. Pallé, *et al.* incl. **C. Cifuentes** SP WR (17/56)
Characterization of two new transiting sub-Neptunes and a terrestrial planet around M-dwarf hosts.
A&A, 707, A106.

[49] A. Abreu, J. Lillo-Box, A. M. Pérez-García, *et al.* incl. **C. Cifuentes** DA WR *Gaia* (6/6)
ExoDNN: boosting exoplanet detection with artificial intelligence. Application to Gaia *Data Release 3.*
A&A, 704, A150 (2025).

[48] J. Sanz-Forcada, E. González-Álvarez, M. R. Zapatero Osorio, *et al.* incl. **C. Cifuentes** WR (18/32)
A planetary system with a sub-Neptune planet in the habitable zone of TOI-2093.
A&A, 703, A93.

[47] S. Geraldía-González, J. Orell-Miquel, E. Pallé, *et al.* incl. **C. Cifuentes** SP WR (16/33)
Discovery of a transiting hot water-world candidate orbiting Ross 176 with TESS and CARMENES.
A&A, 700, A216.

[46] H. L. Ruh, M. Zechmeister, A. Reiners, *et al.* incl. **C. Cifuentes** SP WR (6/22)
The CARMENES search for exoplanets around M dwarfs. The impact of rotation and magnetic fields on the radial velocity jitter in cool stars.
A&A, 692, A138.

[45] J. González-Payo, J. A. Caballero, J. Gorgas, *et al.* incl. **C. Cifuentes** SP WR (6/6)
Multiplicity of stars with planets in the solar neighbourhood.
A&A, 689, A302.

[44] A. von Stauffenberg, T. Trifonov, A. Quirrenbach, *et al.* incl. **C. Cifuentes** SP WR (15/21)
The CARMENES search for exoplanets around M dwarfs. Revisiting the GJ 581 multi-planetary system with

new Doppler measurements from CARMENES, HARPS, and HIRES.
A&A, 688, A112.

[43] P. Mas-Buitrago, A. González-Marcos, E. Solano, *et al.* incl. **C. Cifuentes** (WR) (12/12)
Using autoencoders and deep transfer learning to determine the stellar parameters of 286 CARMENES M dwarfs.
A&A, 687, A205.

[42] E. Goffo, P. Chaturvedi, F. Murgas, *et al.* incl. **C. Cifuentes** (SP) (WR) (22/50)
TOI-4438 b: a transiting mini-Neptune amenable to atmospheric characterization.
A&A, 685, A147.

[41] S. Dreizler, R. Luque, I. Ribas, *et al.* incl. **C. Cifuentes** (SP) (WR) (15/30)
Teegarden's Star revisited: A nearby planetary system with at least three planets.
A&A, 685, A117.

[40] F. Llorente de Andrés, R. de la Reza, P. Cruz, *et al.* incl. **C. Cifuentes** (SP) (WR) (7/7)
The evolution of lithium in FGK dwarf stars. Influence of planets and Galactic migration.
A&A, 684, A28.

[39] Y. Shan, D. Revilla, S. L. Skrzypinski, *et al.* incl. **C. Cifuentes** (SP) (WR) (6/43)
CARMENES input catalog of M dwarfs: VII. New rotation periods for the survey stars and their correlations with stellar activity.
A&A, 684, A9.

[38] M. Mallorquín, E. Goffo, E. Pallé, *et al.* incl. **C. Cifuentes** (SP) (WR) (34/81)
TOI-1801 b: a temperate mini-Neptune around a young M0.5 dwarf.
A&A, 680, A76.

[37] P. Gorrini, J. Kemmer, S. Dreizler, *et al.* incl. **C. Cifuentes** (SP) (WR) (17/39)
Planetary companions orbiting the M dwarfs GJ 724 and GJ 3988: A CARMENES and IRD collaboration.
A&A, 680, A28.

[36] E. Pallé, J. Orell-Miquel, M. Brady, *et al.* incl. **C. Cifuentes** (SP) (WR) (28/66)
GJ 806 (TOI-4481): A bright nearby multi-planetary system with a transiting hot, low-density super-Earth.
A&A, 678, A80.

[35] J. Orell-Miquel, M. Lampón, M. López-Puertas, *et al.* incl. **C. Cifuentes** (SP) (WR) (17/34)
Confirmation of an He I evaporating atmosphere around the 650-Myr-old sub-Neptune HD 235088 b (TOI-1430 b) with CARMENES.
A&A, 677, A56.

[34] E. González-Álvarez, M. R. Zapatero Osorio, V. J. S. Béjar, **C. Cifuentes** (SP) (WR), *et al.* (5/53)
Two sub-Neptunes around the M dwarf TOI-1470.
A&A, 675, A177.

[33] E. González-Álvarez, J. Kemmer, P. Chaturvedi, *et al.* incl. **C. Cifuentes** (SP) (WR) (8/32)
The CARMENES search for exoplanets around M dwarfs. A sub-Neptunian mass planet in the habitable zone of HN Lib.
A&A, 675, A141.

[32] A. Bello-García, V. M. Passegger, J. Ordieres-Meré, *et al.* incl. **C. Cifuentes** SP WR (12/20)
The CARMENES search for exoplanets around M dwarfs. A deep transfer learning method to determine $T_{\rm eff}$ and [M/H] of target stars.
A&A, 673, A105.

[31] R. de la Reza, C. Chavero, S. Roca-Fábrega, *et al.* incl. **C. Cifuentes** SP WR DA (6/6)
Searching for the nature of stars with debris disks and planets.
A&A, 671, A136.

[30] J. Blanco-Pozo, M. Perger, M. Damasso, *et al.* incl. **C. Cifuentes** SP WR (8/41)
The CARMENES search for exoplanets around M dwarfs. A long-period planet around GJ 1151 measured with CARMENES and HARPS-N data.
A&A, 671, A50.

[29] I. Ribas, A. Reiners, M. Zechmeister, *et al.* incl. **C. Cifuentes** SP WR DA (22/105)
The CARMENES search for exoplanets around M dwarfs. Guaranteed Time Observations Data Release 1 (2016–2020).
A&A, 670, A139.

[28] D. Kossakowski, M. Kürster, T. Trifonov, *et al.* incl. **C. Cifuentes** SP WR (25/50)
The CARMENES search for exoplanets around M dwarfs. Wolf 1069 b: Earth-mass planet in the habitable zone of a nearby, very low-mass star.
A&A, 670, A84.

[27] M. Damasso, M. Perger, J. M. Almenara, *et al.* incl. **C. Cifuentes** SP WR (19/34)
A quarter century of spectroscopic monitoring of the nearby M dwarf Gl 514. A super-Earth on an eccentric orbit moving in and out of the habitable zone.
A&A, 666, A187.

[26] P. Chaturvedi, P. Bluhm, E. Nagel, *et al.* incl. **C. Cifuentes** SP WR (16/60)
TOI-1468: A system of two transiting planets, a super-Earth and a mini-Neptune, on opposite sides of the radius valley.
A&A, 666, A155.

[25] R. Luque, G. Nowak, T. Hirano, *et al.* incl. **C. Cifuentes** SP WR (11/48)
Precise mass determination for the keystone sub-Neptune planet transiting the mid-type M dwarf G 9-40.
A&A, 666, A154.

[24] D. Kossakowski, M. Kürster, Th. Henning, *et al.* incl. **C. Cifuentes** SP WR (20/37)
The CARMENES search for exoplanets around M dwarfs. Stable radial-velocity variations at the rotation period of AD Leonis: A test case study of current limitations to treating stellar activity.
A&A, 666, A143.

[23] S. V. Jeffers, J. R. Barnes, P. Schöfer, *et al.* incl. **C. Cifuentes** SP WR (13/20)
The CARMENES search for exoplanets around M dwarfs: Benchmarking the impact of activity in high-precision radial velocity measurements.
A&A, 663, A27.

[22] N. Espinoza, E. Pallé, J. Kemmer, R. Luque, J. A. Caballero, **C. Cifuentes** (SP) (WR), *et al.* (6/69)
A Transiting, Temperate Mini-Neptune Orbiting the M Dwarf TOI-1759 Unveiled by TESS.
AJ, 163, 133.

[21] J. Kemmer, S. Dreizler, D. Kossakowski, *et al.* incl. **C. Cifuentes** (SP) (WR) (29/67)
Discovery and mass measurement of the hot, transiting, Earth-sized planet, GJ 3929 b.
A&A, 659, A17.

[20] E. Marfil, H. M. Tabernero, D. Montes, *et al.* incl. **C. Cifuentes** (SP) (WR) (14/27)
The CARMENES search for exoplanets around M dwarfs. Stellar atmospheric parameters of target stars with `SteParSyn`.
A&A, 656, A162.

[19] D. Kossakowski, J. Kemmer, P. Bluhm, *et al.* incl. **C. Cifuentes** (SP) (WR) (23/66)
TOI-1201 b: A mini-Neptune transiting a bright and moderately young M dwarf.
A&A, 656, A124.

[18] S. Roca-Fábrega, F. Llorente de Andrés, C. Chavero, **C. Cifuentes** (WR) (DA), R. de la Reza
The bi-modal ^{7}Li distribution of the Milky Way's thin-disk dwarf stars. The role of Galactic-scale events and stellar evolution.
A&A, 656, A24.

[17] F. Llorente de Andrés, C. Chavero, R. de la Reza, S. Roca-Fábrega, **C. Cifuentes** (WR) (DA)
The evolution of lithium in FGK dwarf stars: The Li rotation connection and the Li desert.
A&A, 654, A137.

[16] V. Perdelwitz, M. Mittag, L. Tal-Or, *et al.* incl. **C. Cifuentes** (WR) (DA) (14/18)
CARMENES input catalog of M dwarfs VI. A time-resolved Ca II H&K catalog from archival data.
A&A, 652, A116.

[15] P. J. Amado, F. F. Bauer, C. Rodríguez López, *et al.* incl. **C. Cifuentes** (SP) (WR) (23/45)
The CARMENES search for exoplanets around M dwarfs. Two terrestrial planets orbiting G 264-012 and one terrestrial planet orbiting Gl 393.
A&A, 650, A188.

[14] T. Trifonov, J. A. Caballero, J. C. Morales, *et al.* incl. **C. Cifuentes** (SP) (WR) (20/69)
A nearby transiting rocky exoplanet that is suitable for atmospheric investigation.
Science, 371, 6533, 1038–1041.

[13] S. Dreizler, M. Crossfield, D. Kossakowski, *et al.* incl. **C. Cifuentes** (SP) (WR) (47/121)
The CARMENES search for exoplanets around M dwarfs. LP 714-47b (TOI 442.01): Populating the Neptune desert.
A&A, 644, A127.

[12] J. Kemmer, S. Stock, D. Kossakowski, *et al.* incl. **C. Cifuentes** (SP) (WR) (36/96)
Discovery of a hot, transiting, Earth-sized planet and a second temperate, non-transiting planet around the M4 dwarf GJ 3473 (TOI-488).
A&A, 642, A236.

[11] G. Nowak, R. Luque, H. Parviainen, *et al.* incl. **C. Cifuentes** (SP) (WR) (12/100)
The CARMENES search for exoplanets around M dwarfs. Two planets on opposite sides of the radius gap transiting the nearby M dwarf LTT 3780.
A&A, 642, A173.

[10] D. Baroch, J. C. Morales, I. Ribas, *et al.* incl. **C. Cifuentes** (WR) (13/29)
The CARMENES search for exoplanets around M dwarfs. Convective shift and starspot constraints from chromatic radial velocities.
A&A, 641, A69.

[9] P. Bluhm, R. Luque, N. Espinoza, *et al.* incl. **C. Cifuentes** (SP) (WR) (25/84)
Precise mass and radius of a transiting super-Earth planet orbiting the M dwarf TOI-1235: a planet in the radius gap?
A&A, 639, A132.

[8] E. Pallé, S. Nortmann, N. Casasayas-Barris, *et al.* incl. **C. Cifuentes** (SP) (WR) (14/27)
A He I upper atmosphere around the warm Neptune in GJ 3470 b.
A&A, 638, A61.

[7] J. C. Morales, A. J. Mustill, I. Ribas, *et al.* incl. **C. Cifuentes** (SP) (WR) (70/182)
A giant exoplanet around a very low mass star challenging formation models.
Science, 365, 6460, 1441–1445.

[6] M. Zechmeister, S. Dreizler, I. Ribas, *et al.* incl. **C. Cifuentes** (SP) (WR) (48/183)
The CARMENES search for exoplanets around M dwarfs. Two temperate Earth-mass planet candidates around Teegarden's star.
A&A, 627, A49.

[5] R. González-Peinado, J. A. Caballero, D. Montes, **C. Cifuentes** (WR) (DA)
Cool dwarfs in wide multiple systems: A curious quintuple system of a compact sun-like triple and a close M dwarf–white dwarf pair at a wide separation.
The Observatory, 138, 292–298.

[4] A. Kaminski, T. Trifonov, J. A. Caballero, *et al.* incl. **C. Cifuentes** (SP) (WR) (17/40)
The CARMENES search for exoplanets around M dwarfs. A Neptune-mass planet traversing the habitable zone around HD 180617.
A&A, 618, A115.

[3] A. Reiners, M. Zechmeister, J. A. Caballero, *et al.* incl. **C. Cifuentes** (WR) (36/171)
The CARMENES search for exoplanets around M dwarfs. High-resolution optical and near-infrared spectroscopy of 324 survey stars.
A&A, 612, A49.

[2] T. Trifonov, M. Kürster, M. Zechmeister, *et al.* incl. **C. Cifuentes** (WR) (54/169)
The CARMENES search for exoplanets around M dwarfs. First visual-channel radial-velocity measurements and orbital parameter updates of seven M-dwarf planetary systems.
A&A, 609, A117.

[1] A. Reiners, I. Ribas, M. Zechmeister, *et al.* incl. **C. Cifuentes** (WR) (45/172)

The CARMENES search for exoplanets around M dwarfs. HD 147379 b: A nearby Neptune in an early-M dwarf's temperate zone.
A&A, 609, A5.

1.3 Refereed contributions to international meetings

[18] J. González-Payo+ Multiplicity of stars with planets (10)
[17] J. A. Caballero+ ARToFCOSMOS: aprendiendo a comunicar la astronomía al público (10)
[16] J. A. Caballero+ Carmencita: Know thy star, know thy planet (10)
[15] **C. Cifuentes**+ About the multiplicity of M dwarfs (9)
[14] **C. Cifuentes**+ Music and astronomy. IV. The Astrophysical Brothers (9)
[13] Y. Shan+ New rotation periods for a very well-characterised sample of nearby M dwarfs (8)
[12] M. Cortés-Contreras+ Kinematics, activity and youth in the CARMENES input catalogue (8)
[11] **C. Cifuentes**+ Multiplicity of M dwarfs in the CARMENES catalogue (7)
[10] J. A. Caballero+ How does Gaia help to the CARMENES exoplanet survey? (6)
[9] J. A. Caballero+ News from Gl 486 b (6)
[8] **C. Cifuentes+** Luminosities of cool stars (5)
[7] Y. Shan+ CARMENES and the Frontiers of High-Resolution Spectroscopy for M dwarfs (3)
[6] E. Marfil+ Stellar atmospheric parameters of CARMENES GTO M dwarfs with `SteParSyn` (4)
[5] Y. Shan+ CARMENES and the Frontiers of High-Resolution Spectroscopy for M dwarfs (3)
[4] M. Cortés-Contreras+ Kinematics of M dwarfs in the CARMENES input catalogue (2)
[3] S. Roca-Fábrega+ The bimodal A(Li) distribution of Milky Way's thin disk stars (2)
[2] **C. Cifuentes**+ Colours and luminosities of M dwarfs in the CARMENES input catalogue (2)
[1] **C. Cifuentes**+ SEDs and luminosities of M dwarfs in the CARMENES search for exoplanets (1)

References for the events:

(1) SEA XIII (Salamanca, 2018)
(2) SEA XIV.0 (*online*, 2020)
(3) PLATO Mission Conference (*online*, 2021)
(4) CS 20.5 – Cool Stars Workshop (*online*, 2021)
(5) The Star–Planet Connection Workshop (*online*, 2021)
(6) EAS Annual Meeting (Valencia, 2022)
(7) CS 21 – Cool Stars Workshop (Toulouse, 2022)
(8) EAS Annual Meeting (Krakow, 2023)
(9) SEA XV (Tenerife, 2022)
(10) SEA XVI (Granada, 2024)

1.4 SPIE conference proceedings

[2] A. Quirrenbach, and the CARMENES Consortium (incl. **C. Cifuentes**).
The CARMENES M-dwarf planet survey
Proceedings of the SPIE, Vol. 11447, id. 114473C 11 pp. (2020).

[1] A. Quirrenbach, and the CARMENES Consortium (incl. **C. Cifuentes**).
CARMENES: high-resolution spectra and precise radial velocities in the red and infrared.
Proceedings of the SPIE, Vol. 10702, id. 107020W.

1.5 Circulars

[1] P. Bacci, M. Maestripieri, M. Facchini, *et al.* (incl. **C. Cifuentes**).

(Observer of target of opportunity at the William Herschel Telescope)
MPEC 2020-A99: 2020 AV2.
Minor Planet Electronic Circulars, No. 2020-A99. January 2020.

1.6 Catalogues

[43] González-Payo+ Known multiple stellar systems within 10 pc J/MNRAS/549/G838
[42] Varas+ CARMENES M dwarfs projected rotational velocities J/A+A/709/A244
[41] Poultourtzidis+ TOI-1243, TOI-4529, TOI-5388 RV datasets J/A+A/707/A106
[40] Abreu+ ExoDNN v1 candidates J/A+A/704/A150
[39] Geraldía-González+ Ross 176 photometry, RVs, activity indices J/A+A/700/A216
[38] Balsalobre-Ruza+ KOBE-1 activity indicators and RVs J/A+A/694/A15
[37] **Cifuentes+** CARMENES input catalogue of M dwarfs. IX J/A+A/693/A228
[36] Cortés-Contreras+ CARMENES input catalogue of M dwarfs. VIII J/A+A/692/A206
[35] Ruh+ Radial velocity jitter in cool stars J/A+A/692/A138
[34] González-Payo+ Multiplicity of stars with planets J/A+A/689/A302
[33] von Stauffenberg+ GJ 581 RVs and activity indices J/A+A/688/A112
[32] Mas-Buitrago+ Stellar parameters of 286 CARMENES M dwarfs J/A+A/687/A205
[31] Dreizler+ Stellar parameters of 286 CARMENES M dwarfs J/A+A/684/A117
[30] Llorente de Andrés+ Stellar parameters of 286 CARMENES M dwarfs J/A+A/684/A28
[29] Shan+ Stellar parameters of 286 CARMENES M dwarfs J/A+A/684/A9
[28] González-Álvarez+ HN Lib radial velocities J/A+A/675/A141
[27] Gorrini+ GJ 724 and GJ 3988 RV timeseries J/A+A/680/A28
[26] Bello-García+ Deep Transfer Learning of $T_{\rm eff}$ and [M/H] J/A+A/673/A105
[25] de la Reza+ Stars with debris disks and planets J/A+A/671/A136
[24] Blanco-Pozo+ GJ 1151 CARMENES and HARPS-N data J/A+A/671/A50
[23] Kossakowski+ Wolf 1069 RV and stellar activity indices J/A+A/670/A84
[22] Ribas+ CARMENES search for exoplanets around M dwarfs J/A+A/670/A139
[21] Kossakowski+ AD Leo RV and stellar activity indices J/A+A/666/A143
[20] Chaturvedi+ TOI-1468 photometry and radial velocities J/A+A/666/A155
[19] Espinoza+ VRI photometry and radial velocity of TOI-1759 J/AJ/163/133
[18] Damasso+ Gl 514 RVs and Activity diagnostics J/A+A/666/A187
[17] Kemmer+ GJ 3929 b RVs and activity indicators J/A+A/659/A17
[16] Marfil+ CARMENES stellar atmospheric parameters J/A+A/656/A162
[15] Kossakowski+ TOI-1201 RV and activity index J/A+A/656/A124
[14] Llorente de Andrés+ Evolution of Li in FGK dwarf stars J/A+A/654/A137
[13] Perdelwitz+ CARMENES time-resolved CaII H&K catalog J/A+A/652/A116
[12] Amado+ G 264-012 and Gl 393 radial velocity curves J/A+A/650/A188
[11] Martínez-Rodríguez+ Compilation of planets around M dwarfs J/ApJ/887/261
[10] Dreizler+ LP 714-47 (TOI 442) radial velocity curve J/A+A/644/A127
[9] Kemmer+ GJ 3473 (TOI-488) radial velocity curve J/A+A/642/A236
[8] Morales+ GJ 3512 radial velocity and light curves J/other/Sci/365.1441
[7] **Cifuentes+** CARMENES input catalogue of M dwarfs. V J/A+A/642/A115
[6] Zechmeister+ Teegarden's Star RV and Hα curves J/A+A/627/A49
[5] Schweitzer+ Radii and masses of the CARMENES targets J/A+A/625/A68
[4] Kaminski+ A Neptune-mass planet traversing the habitable zone around HD 180617 J/A+A/618/A115
[3] Reiners+ 324 CARMENES M dwarfs velocities J/A+A/612/A49
[2] Reiners+ HD147379 b velocity curve J/A+A/609/L5
[1] Trifonov+ CARMENES radial velocity curves of 7 M-dwarf J/A+A/609/A117

1.7 Book chapters and sections

[1] D. Barrado Navascués and S. Cabañero, and 50+ coauthors. *100 Conceptos de Astrobiología* Ed. INTA-CSIC (2022)

2 Observational experience

HARPS ESO 3.6m

Triple Trouble: Inner planets in hierarchical triple systems
Cifuentes | 2026A | 51 h | Completed

The atmosphere of a brave dying world
Lillo-Box | 2025B | 12 h | Completed

CARMENES CAHA 3.5m

Triple Trouble: Planets in triple star systems
Cifuentes | 2026B | 1 n | service | Observing

Triple Trouble: Inner planets in hierarchical triple systems
Cifuentes | 2026A | 1 n | service | Completed

Triple Trouble: Inner planets in hierarchical triple systems
Cifuentes | 2025A | 2 n | service | Completed

Calibrating abundances of M dwarfs with wide visual binaries
Duque-Arribas | 2023A | 2.9n | 2023B | 1.20 n | 2024A | 2.25 n | 2024B | 3.60 n | Completed

CAFE CAHA 2.2m

Make CAFE great again: confirming left-behind transiting exoplanet candidates
Lillo-Box | 2025A | 9 n | Completed

AstraLux CAHA 2.2m

Getting lucky for KOBE
Cifuentes | 2026A | 4 n | service | Completed

HARPS-N TNG

The HARPS-N contribution to the KOBE survey
Lillo-Box | 2025B | 6 n | Observing

FIES NOT

Identifying new M-dwarf spectroscopic binaries from Gaia RV data
Cifuentes | 2026A | 2 n | observer | Completed
Cifuentes | 2022B | 4 n | observer | Completed
Cifuentes | 2023A | 4 n | observer | Completed

SPIRou CFHT

NIR confirmation of a planet around a late-K dwarf

Santerne | 2024B | 2 h | DDT | Completed

HERMES MERCATOR

Calibrating abundances of M dwarfs with wide visual binaries

Duque-Arribas | 2022B | 3.0 n | Completed

SES STELLA

Calibrating abundances of M dwarfs with wide visual binaries

Duque-Arribas | 2022B | 0.33 n | Completed

ESPRESSO VLT

Brown dwarf candidate characterisation

Zapatero Osorio | 2021B | 1 n | remote | Completed

WFC INT

Formation origins of brown dwarfs from rotation rates

Lodieu | 2020B | 5 n | observer | Completed

NGTS Paranal

Characterizing the pulsations of a planet-host candidate sub-giant star

Cifuentes | 2026A | service | Completed

ISIS WHT

Spectroscopy of poorly investigated late-type young stars in the σ Ori cluster

Cifuentes | 2019B | 4.5 n | observer | Completed

3 Research funding

Postdoctoral Researcher

Centro de Astrobiología (CAB-CSIC) | 2024–2026

- **Ramón y Cajal programme**
  RYC2021-031640-I | PI: J. Lillo-Box
- Associated to the **Spanish National Project**
  PID2023-150468NB-I00 | PIs: D. Barrado, N. Huélamo
- Additional institutional support from the **CSIC Intramural Special Project** PIE 2023AT003 and the **María de Maeztu Excellence Unit** MDM-2017-0737

Predoctoral Researcher

Centro de Astrobiología (CAB-CSIC) | 2018–2022

- **Formación de Personal Investigador** (FPI)
  AYA2016-79425-C3-2-P | PIs: M. R. Zapatero Osorio, J. Sanz Forcada
- Associated to the **Spanish National Project**
  PID2019-109522GB-C51 | PIs: M. R. Zapatero Osorio, J. Sanz Forcada

4 Scientific meetings

Symbols indicate the type of contribution: invited/oral (●) and poster (○).

4.1 National and international congresses

Dec 2025 Exploring Tatooine and Beyond: Circumbinary Planets with ESA Missions, Madrid, Spain
Oct 2025 ESA's Madrid-Area Exoplanet Science Meeting (MAESM) 2025, Madrid, Spain ●
Jul 2025 Detection and Dynamics of Exoplanets (DDE), Coimbra, Portugal ●
Oct 2022 ESA's Madrid-Area Exoplanet Science Meeting (MAESM) 2022, Madrid, Spain
Sep 2022 Spanish Astronomical Society XV Meeting, Tenerife, Spain ● ○
Jun 2022 Cool Stars 21, Toulouse, France ○
Oct 2021 The Star-Planet Connection, *online*, ESO ○
May 2021 CAP 2021: Communicating Astronomy with the Public, *online*, IAU ○
Mar 2021 Cool Stars 20.5 Virtual Meeting, *online* ○ ○
Jul 2020 Spanish Astronomical Society XIV.0 Meeting, *online* ○ ○ ○
Jul 2020 European Astronomical Society Annual Meeting, Granada, Spain ○
Feb 2019 1st meeting of the Spanish exoplanet network (Exonet), Granada, Spain ●
Jul 2018 Spanish Astronomical Society XIII Meeting, Salamanca, Spain ○ ○ ○

4.2 Consortia & Project meetings

Mar 2026 CARMENES 23th Scientific Meeting, Madrid, Spain
Nov 2025 KOBE Hackathon Week #4, *online*
Mar 2025 KOBE Hackathon Week #3, Geneva, Switzerland
Nov 2024 KOBE Hackathon Week #2, *online*
Nov 2024 CARMENES 20th Scientific Meeting, *online* ●
Jun 2022 CARMENES 16th Scientific Meeting, Göttingen, Germany ●
Nov 2021 CARMENES 15th Scientific Meeting, *online*
May 2021 CARMENES 14th Scientific Meeting, *online*
Nov 2020 CARMENES 13th Scientific Meeting, *online*
Mar 2020 CARMENES 12th Scientific Meeting, *online*
Nov 2019 CARMENES 11th Scientific Meeting, Weimar, Germany ●
May 2019 CARMENES 10th Scientific Meeting, Seville, Spain ●
May 2019 CARMENES 9th Scientific Meeting, Barcelona, Spain ●
Apr 2017 CARMENES 6th Scientific Meeting, Madrid, Spain

4.3 Invited talks

Feb 2026 M dwarfs with exoplanets Lund, Sweden

4.4 Schools, Workshops & Summer Courses

Nov 2023 Writing Better Research Papers and Proposals (A. Hatzes) *Online*
Sep 2020 Scientific Writing and Oral Presentations (IAA) Madrid, Spain
Jan 2020 I Jornada de Invierno de Astrobiología Madrid, Spain
Dec 2019 Machine Learning and Applications to Physics (IPARCOS) Madrid, Spain
Jul 2019 Cursos de Verano de El Escorial: 20 Años de Astrobiología en España Madrid, Spain
Jun 2018 *Gaia* DR2 Exploration Lab Madrid, Spain
May 2018 Virtual Observatory 11th School Madrid, Spain
May 2018 Deep Learning as a Tool for Scientific Data Analysis Madrid, Spain

Dec 2017 Aspectos Metodológicos para el Doctorado Madrid, Spain
Nov 2017 ASTERICS VO School: Data Access, Discovery and Interoperability (DADI) Madrid, Spain

4.5 Other

Selected informal presentations

Apr 2026 Paper Review: Ávila-Bravo *et al.* (2025) (Journal Club), Madrid, Spain ●
Feb 2026 Use ADS with Python (PyCoffee), Madrid, Spain ●
May 2025 Paper Review: Tokovinin *et al.* (2021) (Journal Club), Madrid, Spain ●
Oct 2024 Astroquery and its secrets (PyCoffee), Madrid, Spain ●

Poster contributions as a co-author (no attendance)

Jun 2026 Cool Stars 23, Tokyo, Japan ○
Jul 2024 Spanish Astronomical Society XVI Meeting, Granada, Spain ○○○
Jul 2022 European Astronomical Society Meeting 2022, Valencia, Spain ○○
Oct 2021 The Star-Planet Connection workshop, *online* ○
Aug 2021 TESS Science Conference 2 (TSC2), *online* ○○○○
Jul 2020 European Astronomical Society Meeting 2020, *online* ○○○
Sep 2017 Annual Meeting of the Astronomische Gesellschaft 2017, Göttingen, Germany ○○

Predoctoral events

Oct 2020 II Jornadas para Doctorandos del CSIC, *online* ●
Jun 2019 I Jornadas para Doctorandos del CSIC, *online* ●
Apr 2019 Concurso "Tesis en 3 minutos", Madrid, Spain ●
Mar 2019 Jornadas de Doctorandos UCM, Madrid, Spain
Nov 2018 PhDay 2018, Madrid, Spain ● ○
Dec 2017 PhDay 2017, Madrid, Spain

Miscellanea & Music

Oct 2023 Creation of audiovisuals for two presentations at ESA Open Day 2023 ESAC, Madrid
Aug 2022 Collaboration on the radio programme *El Universo Cultural Círculos Concéntricos*, RNE
Jun 2022 Piano and bass guitar performance at ESA Day 2022 ESAC, Madrid
Jun 2021 Piano and audiovisual production for Fête de la Musique 2021 ESAC, Madrid
2018 & 2022 Playlist curation for Astromúsica Cultura con C de Cosmos (CSIC–INTA)
2025–today Electronica, metal and instrumental personal projects (ccifuentesr.io/musik).

5 Service

5.1 Refereeing

[2] Anonymous referee for *Astronomy & Astrophysics*.
[1] Public referee for *Qeios* journal.

5.2 Local Organising Committee

Dec 2025 Exploring Tattoine and Beyond: Circumbinary Planets with ESA Missions ESAC, Madrid

5.3 Outreach & Volunteering

Oct 2023 Reception at the 2nd ESA Open Day, ESAC
Mar 2022 Support at Diálogos en el Café Gijón, Cultura con C de Cosmos (CSIC-INTA)
Oct 2019 Reception and solar observation at the 1st ESA Open Day, ESAC
2016–2017 Class representative in Master in Astrophysics course, UCM

6 Academic background

6.1 Degrees

2018–2023 *PhD* in **Astrophysics** Universidad Complutense de Madrid
International Doctorate Mention, *summa cum laude*
Thesis: Astrophysical parameters of M dwarfs with exoplanets
Advisors: Dr. José Antonio Caballero, Dr. Jorge Sanz Forcada

2016–2017 *MSc* in **Astrophysics** Universidad Complutense de Madrid
Thesis: Characterisation of the CARMENES sample: Carmencita photometry update and calculation of luminosities
Advisors: Dr. José Antonio Caballero, Prof. David Montes

Solar System and exoplanets • *Cool stars and substellar objects* • *Stellar atmospheres* • *Interstellar medium* • *Galactic formation and evolution* • *Astronomical instrumentation* • *Experimental techniques in astrophysics* • *Data analysis and statistical techniques.* (60 ECTS)

2013–2014 *MSc* in **Renewable Energies** Universidad Autónoma de Madrid
Thesis: The effect of heat treatment on the sintering of spinel-like compounds for solar selective absorbers.
Advisors: Dr. Ángel Morales Sabio, Dr. Nuria Germán Cordero

Wind and Solar power • *Hydropower* • *Geothermal Energy* • *Biomass* • *Nuclear Energy: Fission and Fusion* • *Energy Storage: Hydrogen, Fuel Cells* • *Energy conversion, transportation and storage* • *Bioclimatic architecture* • *Environment, energy economy and sustainable development.* (60 ECTS)

2007–2013 *BSc / Licentiate* in **Physics** Universidad de Salamanca

Specialization on Astrophysics and Earth Sciences: Astrophysics • *Climatology* • *Meteorology* • *Atmospheric thermodynamics and radiation* • *Photonics and electronics.* (314 ECTS)

6.2 Complementary courses

Programming, Data Science and OS

Aug 2026 Deep Learning with PyTorch, Keras and Tensorflow, IBM, *online*
Dec 2018 Data Science: Productivity Tools, HarvardX, *online*
Dec 2017 Data Driven Astronomy with Python and SQL, University of Sidney, *online*
Mar 2011 Linux, Escuela Julián Besteiro, *online*
Jun 2009 Scientific Software in Mathematics: Mathematica and MATLAB, USAL, Salamanca

Astronomy

Mar 2016 Cosmology, Australian National University, *online*
Nov 2015 Philosophy and the Sciences, University of Edinburgh, *online*
Jul 2015 The Science & Technology behind Astronomical Discovery, University of Edinburgh, *online*

Languages

Nov 2020 Inglés específico: presentaciones C1, Euroinnova, *online*, 30h

2011–2012 Upper-Intermediate Course in English, Servicio Central de Idiomas USAL, Salamanca, 60h
Oct 2009 English immersion course, UIMP, Tenerife, 40h
Jun 2010 English immersion course, UIMP, Barcelona, 40h

Miscellanea

Apr 2018 Scopus, FECYT, *online*, 2h
Mar 2015 Critical Thinking in Global Challenges, University of Edinburgh, *online*
Dec 2014 Wind Power: Fundamentals, technology and applications, CIEMAT, *online*, 50h
Dec 2014 Learning How to Learn, University of California, San Diego, *online*
Sep 2014 Fundamentals of Music Theory, University of Edinburgh, *online*

7 Skills

Computational

Beginner means *used at some point or occasionally*;
Operational means *extensive experience with*;
Proficient means *day-to-day comfort.*

	Beginner	Operational	Proficient
Programming	`C`	Bash	`Python`
Machine learning	`PyTorch, Keras, Tensorflow`	`scikit-learn`	
Scientific Python		`Numpy, Pandas, SciPy`	
Mathematics & Statistics	MATLAB	Mathematica, R	
Databases	`postgreSQL`	`SQL, ADQL)`	
Version control		Git	
Astronomy (Databases)			CDS(1)
Astronomy (Exoplanet)	`exoplanet`, `juliet`	`exostriker`	
Astronomy (Spectra)	`IRAF, DS9`		`iSpec`
Document preparation		Markdown	LaTeX(2)
Web & design	`jQuery`	`HTML, CSS`	
Operating Systems		Linux, Windows	macOS

(1) CDS (Centre de Données astronomiques de Strasbourg) includes ADS, Simbad, VizieR, and Aladin.
(2) Additionally, office suites such as Microsoft *Office* (Word, PowerPoint, Excel) and macOS *iWork* (Pages, Keynote, Numbers). Touch typing of ∼50 wpm.

8 Miscellanea

Additional experience

2015 Private lessons (Mathematics, High school level)

2015–2016 Telephone operator (*teleoperador*) in regional and national Spanish elections.

Affiliations

2024+ International Astronomical Union (IAU)
Division A: Fundamental Astronomy - **Division F:** Planetary Systems and Astrobiology - **Division G:** Stars and Stellar Physics - Executive Committee WG Junior Members

2021–2026 European Astronomical Society (EAS)

2018–2026 Sociedad Española de Astronomía (SEA)

Qualifications

2019, 2025 First Aid training (10h)

2025 Fire safety basics training (2h)

2007+ Driving license B

9 Personal details & References

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ADS	Cifuentes C.
Scholar	Cifuentes C.
GitHub	github.com/ccifuentesr
Zenodo	ccifuentesr

References:

- Dr. Jorge Lillo-Box (CAB; jlillo@cab.inta-csic.es, jlillobox.com)
- Dr. José Antonio Caballero Hernández (CAB; caballero@cab.inta-csic.es, exoterrae.eu)
- Dr. Trifon Trifonov (MPIA; trifonov@mpia.de, trifonov.in)
- Prof. Dr. David Montes Gutiérrez (UCM; dmontes@ucm.es)
- Dr. Jorge Sanz Forcada (CAB; jsanz@cab.inta-csic.es)
- Dr. María Rosa Zapatero Osorio (CAB; mosorio@cab.inta-csic.es)
- Dr. Andreas Schweitzer (HS; andreas.schweitzer@hs.uni-hamburg.de)
- Prof. Dr. Taieb Gasmi (SLU; taieb.gasmi@slu.edu)
- Prof. Dr. Ángel Morales Sabio [•] (CIEMAT; amorales@ciemat.es)

CAB: Centro de Astrobiología; CIEMAT: Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas; HS: Hamburger Sternwarte; MPIA: Max-Planck-Institut für Astronomie; SLU: Saint Louis University Madrid; UCM: Universidad Complutense de Madrid.